

Forensics for Cybercrimes Committed Using Artificial Intelligence

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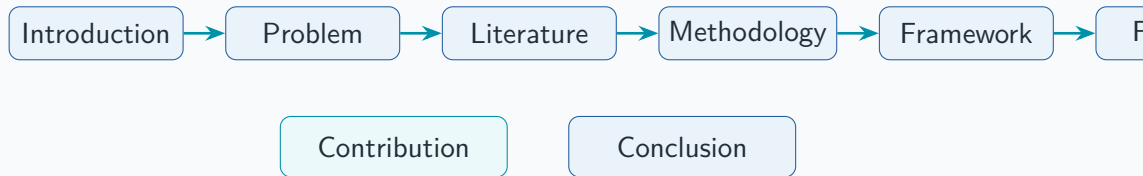
A Digital Forensic Framework for AI-Assisted Cybercrime

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Research Roadmap



Artificial Intelligence is changing the cybercrime landscape.

Traditional Cybercrime

- Human-controlled systems
- Identifiable devices
- Conventional malware
- Established forensic procedures

AI-Assisted Cybercrime

- Automated attacks
- Synthetic identities
- AI-generated content
- Autonomous decision-making

The Forensic Challenge



What happened?



Who or what caused it?



Can we prove it?

Problem Statement

Traditional digital forensic approaches often assume that evidence can be associated with identifiable devices, accounts, users or processes.

AI-assisted cybercrime introduces additional questions:

- Who actually performed the action?
- Was the action human-generated or AI-assisted?
- How can investigators establish provenance?
- Can AI-generated evidence be authenticated?
- How should attribution be established?
- How can such evidence be presented in court?

Research Objectives

General Objective

To develop a digital forensic framework for investigating cybercrimes committed using Artificial Intelligence.

Specific Objectives

1. Examine AI-assisted cybercrime techniques.
2. Evaluate existing digital forensic approaches.
3. Investigate evidence provenance and integrity.
4. Develop a forensic framework.
5. Evaluate the proposed framework.

Research Questions

1. How is AI being used to facilitate cybercrime?
2. What limitations exist in current digital forensic approaches?
3. How can AI-generated evidence be authenticated?
4. How can provenance and attribution be established?
5. What forensic framework can support investigation and courtroom presentation?

Literature Review

Digital Forensics

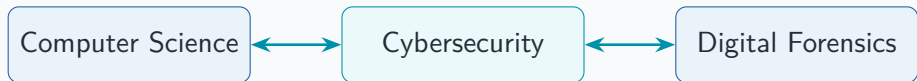
- Evidence acquisition
- Preservation
- Examination
- Analysis
- Presentation

AI and Cybercrime

- Generative AI
- Deepfakes
- Automated attacks
- AI-assisted fraud
- Synthetic content

Theoretical Framework

The research connects computational reasoning with digital forensic analysis.



Formal reasoning provides a mechanism for expressing relationships between evidence, authentication, provenance and attribution.

Formalising Digital Evidence

Let

$$E = \{e_1, e_2, \dots, e_n\}$$

represent the set of collected digital evidence.

Let

$$A \subseteq E$$

represent evidence that has passed the authentication criteria.

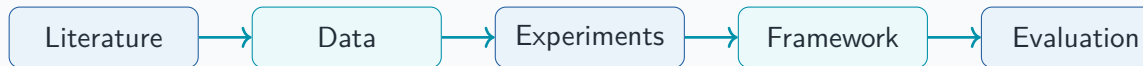
Therefore,

$$\boxed{A \subseteq E}$$

and

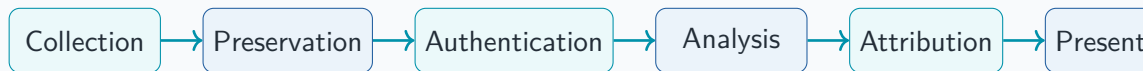
$e_i \in A \Rightarrow e_i$ satisfies the defined authentication criteria.

Research Methodology



- Literature analysis
- Dataset analysis
- Digital forensic experimentation
- AI-assisted cybercrime scenarios
- Framework development
- Experimental evaluation

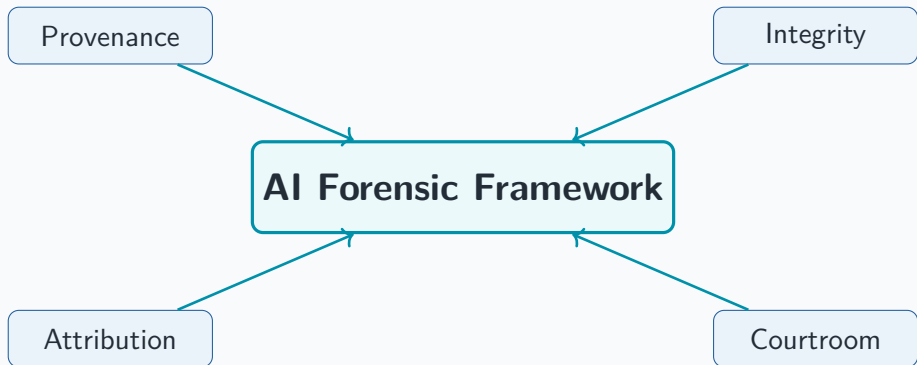
Proposed Forensic Framework



Core Principle

Evidence must remain traceable from its initial collection through analysis and ultimately to its presentation.

Expected Contribution to Knowledge



Conclusion

Central Argument

AI changes not only how cybercrimes can be committed, but also how digital evidence must be interpreted, authenticated and attributed.

AI + Cybercrime + Digital Forensics + Evidence

Thank You

Questions & Discussion

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